

A bounded ultrametric counterexample to the right-semicontinuous uniform-convergence claim

Automatic proof packet for arXiv:2507.05276

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Status. Candidate counterexample, likely valid, pending human review. The example disproves the uniform-on-bounded-sets conclusion of Corollary 3.9 and the uniform clause of Theorem 3.11 in Khandani’s arXiv:2507.05276v1. It does not challenge the pointwise-convergence or fixed-point clauses.

1 Source claim

Khandani’s paper [1] states in Corollary 3.9 that the bounded-set uniform-convergence theorem remains valid when upper semicontinuity of the control function is weakened to upper semicontinuity from the right. The corollary depends on Lemma 3.8, which claims that every positive function on a compact interval that is lower semicontinuous from the right has positive infimum. Theorem 3.11 then incorporates the same uniform-convergence claim under any of three additional fixed-point hypotheses.

The source statements and their proof dependency appear on pages 9–10:

- f is lower semicontinuous from the right on $[a, b]$.

Then

$$\inf_{t \in [a, b]} f(t) > 0.$$

Proof. For each $t \in [a, b]$, since f is lower semicontinuous from the right and positive at t , there exists $\delta_t > 0$ such that

$$f(s) > \frac{f(t)}{2} \quad \text{for all } s \in [t, t + \delta_t] \cap [a, b].$$

The intervals

$$U_t := [t, t + \delta_t] \cap [a, b]$$

form an open cover of the compact interval $[a, b]$. Hence, a finite subcover $\{U_{t_i}\}_{i=1}^N$ exists such that

$$[a, b] \subseteq \bigcup_{i=1}^N U_{t_i}.$$

Define

$$\varepsilon := \min_{1 \leq i \leq N} \frac{f(t_i)}{2} > 0.$$

Then $f(t) > \varepsilon$ for all $t \in [a, b]$, so

$$\inf_{t \in [a, b]} f(t) \geq \varepsilon > 0.$$

□

Corollary 3.9. *The conclusions of Theorems 3.6 and 3.7 remain valid if the assumption that “ ϕ is upper semicontinuous” is replaced by the weaker condition that “ ϕ is upper semicontinuous from the right.”*

Proof. If ϕ is upper semicontinuous from the right, then the function $t \mapsto t - \phi(t)$ is lower semicontinuous from the right. Since $t - \phi(t) > 0$ for all $t > 0$, Lemma 3.8 ensures the positivity of the relevant infimum required in the proof of Theorem 3.6, thus providing the necessary $\delta > 0$. Consequently, the arguments in both Theorems 3.6 and 3.7 remain valid under this relaxed assumption. □

We show that the assumption of a bounded orbit in Theorem 2.5 is superfluous. Furthermore, we demonstrate that the requirement of upper semicontinuity for the control function ϕ can be weakened to right upper semicontinuity without affecting the validity of the result.

Corollary 3.10. *Theorem 2.5 remains valid without assuming that T has a bounded orbit, and if the upper semicontinuity assumption on ϕ is replaced by right upper semicontinuity.*

Proof. By Corollary 3.4, T is a Leader contraction. Hence, the sequence $\{T^n x\}$ converges to some $z \in X$ for all $x \in X$. According to Theorem 3.1 in [11], the existence of such convergence ensures that T has a fixed point. In turn, this implies that T has a bounded orbit. Therefore, all the conditions of

Theorem 2.5 are eventually satisfied, and the conclusion remains valid. Moreover, by Lemma 3.8 and as noted in Corollary 3.9, the proof of Theorem 1.3 in [11] continues to hold when the upper semicontinuity of ϕ is weakened to right upper semicontinuity. Thus, both relaxations are justified. $\square$

Now, we formulate our extension of Kirk's theorem as follows:

Theorem 3.11. *Let (X, Δ) be a b -metric space, $\phi, \phi_n : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, and $T : X \rightarrow X$ satisfy:*

1. $0 < \phi(t) < t$ for all $t > 0$, and ϕ is upper semicontinuous from the right at each $t > 0$.
2. $\phi_n(t) \rightarrow \phi(t)$ uniformly as $n \rightarrow \infty$ on every bounded interval $[0, b]$.
- 3.

$$\Delta(T^n x, T^n y) \leq \phi_n(\Delta(x, y)) \quad \text{for all } x, y \in X, n \in \mathbb{N}.$$

Then $T^n x$ converges to some point $z \in X$ for all $x \in X$. Furthermore, z is the unique fixed point of T , and $T^n x$ converges uniformly to z as $n \rightarrow \infty$ under any of the following conditions:

1. T has a complete graph.
2. There exists $n_* \in \mathbb{N}$ such that $\phi_{n_*}(0) = 0$ and ϕ_{n_*} is upper semicontinuous.
3. T^n is continuous for some $n \in \mathbb{N}$.

Proof. By Lemma 3.2, $T^n x$ converges to some point $z \in X$ for all $x \in X$.

1. Follows from [9, Corollary 4].
2. This condition was introduced by Chen [4, Theorem 2.2]. If $T^n x \rightarrow z$, then

$$\limsup_{n \rightarrow \infty} \Delta(T^{n_*+n} x, T^{n_*} z) \leq \limsup_{n \rightarrow \infty} \phi_{n_*}(\Delta(T^n x, z)) \leq \phi_{n_*}(0) = 0,$$

implying $T^{n_*} z = z$. Also,

$$T^{n_*}(Tz) = T(T^{n_*} z) = Tz,$$

so Tz is a fixed point of T^{n_*} . Since T^{n_*} has a unique fixed point, $Tz = z$.

3. This case follows from [11, Theorem 3.1].

If T has a fixed point, and ϕ is upper semicontinuous from the right at each $t > 0$, then by Corollary 3.9, $T^n x$ converges uniformly to z on bounded subsets of X . $\square$

4. Examples

The following example, presented by Jachymski et al. [6], shows that Kirk-type contractions can be fixed point free in the absence of additional assumptions such as continuity. Since, by Lemma 3.2, every Kirk contraction is also a Leader contraction, this example illustrates the necessity of supplementary conditions—such as continuity or completeness of the graph—to ensure that the limit of the Picard sequence is actually a fixed point. Despite satisfying the assumptions of Lemma 3.2 and being a Leader contraction, the mapping

2 Proof intuition

Use a bounded ultrametric space consisting of countdown chains of every finite length. The point x_k needs exactly k iterates to reach the fixed point x_0 . Distances $r_k = k/(k+1)$ approach the unattained diameter 1. One step lowers r_k to r_{k-1} , so a control function can lie strictly below the identity at every distance. The gaps $r_k - r_{k-1}$ nevertheless tend to zero as $r_k \uparrow 1$. Right semicontinuity at 1 imposes no condition on this left-hand approach. Consequently every orbit converges (indeed terminates), while no common iteration count works for the entire bounded space.

3 The counterexample

Let

$$r_0 = 0, \quad r_k = \frac{k}{k+1} \quad (k \geq 1),$$

and let $X = \{x_k : k \in \mathbb{N}_0\}$. Define

$$d(x_j, x_k) = \begin{cases} 0, & j = k, \\ r_{\max\{j,k\}}, & j \neq k. \end{cases}$$

Define $T : X \rightarrow X$ by

$$Tx_0 = x_0, \quad Tx_k = x_{k-1} \quad (k \geq 1).$$

Finally, define $\phi : [0, \infty) \rightarrow [0, \infty)$ by $\phi(0) = 0$ and

$$\phi(t) = \begin{cases} t/2, & 0 < t < r_1, \\ 1/4, & r_1 \leq t < r_2, \\ r_{k-1}, & r_k \leq t < r_{k+1} \quad (k \geq 2), \\ 1/2, & t \geq 1. \end{cases}$$

For every positive integer n , set $\phi_n = \phi$.

Theorem 1. *The complete metric space (X, d) , map T , and functions $\phi_n = \phi$ satisfy all hypotheses of Theorem 3.11 in [1]. The map T is continuous, has a complete graph and unique fixed point x_0 , and $T^n x$ converges to x_0 for every $x \in X$. However, the convergence is not uniform on the bounded set X .*

Proof. First, d is an ultrametric. Indeed, for three distinct indices, the largest index determines the two largest pairwise distances, which are equal; cases with repeated points are immediate. Every two distinct points are at distance at least $r_1 = 1/2$. Hence every Cauchy sequence is eventually constant, so X is complete. Also $\text{diam } X = 1$, where the diameter is a supremum, so X is bounded. The same separation shows that X is discrete. Therefore T is continuous, its graph is closed and, as a closed subspace of the complete product $X \times X$, its graph is complete.

The function ϕ is positive and strictly below the identity on $(0, \infty)$. This is immediate on $(0, r_1)$ and $[1, \infty)$. On $[r_1, r_2)$, $\phi = 1/4 < r_1$, and on $[r_k, r_{k+1})$ for $k \geq 2$, $\phi = r_{k-1} < r_k$. It is right-continuous at every point except that only the one-sided statement is needed at the step boundaries; directly from the half-open definition, the right limit equals the displayed value. At

$t = 1$ and for $t > 1$ it is constant from the right. Thus ϕ is upper semicontinuous from the right at every positive t . Notice that it is not upper semicontinuous at 1 in the two-sided sense, since

$$\lim_{k \rightarrow \infty} \phi(r_k) = \lim_{k \rightarrow \infty} r_{k-1} = 1 > \phi(1) = \frac{1}{2}.$$

The constant sequence $\phi_n = \phi$ converges uniformly to ϕ on every bounded interval.

It remains to verify the asymptotic contraction inequality. We have

$$T^n x_k = x_{\max\{k-n, 0\}}.$$

For distinct x_j, x_k , suppose without loss of generality that $j < k$. For $n \geq 1$, either both iterates are x_0 , or their distance is $r_{k-n} \leq r_{k-1}$. If $k = 1$, both iterates already equal x_0 ; if $k \geq 2$, then $\phi(d(x_j, x_k)) = \phi(r_k) = r_{k-1}$. Therefore

$$d(T^n x_j, T^n x_k) \leq \phi_n(d(x_j, x_k)) \quad (n \geq 1).$$

The case $j = k$ follows from $\phi_n(0) = 0$. Thus all three numbered hypotheses of source Theorem 3.11 hold.

The only fixed point is x_0 , and each orbit reaches it after finitely many steps. Uniform convergence on X nevertheless fails. For every $n \geq 0$,

$$T^n x_{n+4} = x_4, \quad d(T^n x_{n+4}, x_0) = r_4 = \frac{4}{5}.$$

Hence no iteration index makes all orbit points even $3/4$ -close to x_0 . Equivalently,

$$\sup_{x \in X} d(T^n x, x_0) = 1 \quad \text{for every } n.$$

This contradicts the asserted bounded-set uniform convergence. Since T is continuous and has a complete graph, it satisfies both conditions 1 and 3 listed after the conclusion in source Theorem 3.11. $\square$

4 Why the source proof fails

Lemma 3.8 in [1] is false. For example, on $[1/2, 1]$ let

$$f(t) = 1 - t \quad (1/2 \leq t < 1), \quad f(1) = 1.$$

Then $f > 0$ and f is lower semicontinuous from the right, but $\inf f = 0$. The intervals $[t, t + \delta_t]$ used in the source proof are not open in the usual topology, so compactness does not yield the claimed finite subcover. In the constructed dynamical example, the same obstruction is encoded by

$$r_k - \phi(r_k) = r_k - r_{k-1} \longrightarrow 0.$$

Full upper semicontinuity would prevent this behavior on a compact interval: $t - \phi(t)$ would be positive and lower semicontinuous, hence would have a positive minimum. A weaker repair is to assume directly that $\inf_{t \in [a, b]} (t - \phi(t)) > 0$ for every $0 < a < b < \infty$. The packet does not claim that this is the weakest possible repair.

5 Verification and novelty limits

The accompanying script uses exact rational arithmetic to check all ultrametric triples through index 80, all contraction inequalities for iterates 1 through 40 on that truncation, and the explicit witnesses x_{n+4} for $0 \leq n \leq 40$. These tests check transcription and indexing; the proof above establishes the infinite statements.

A bounded literature search on 9 August 2026 used the exact source title and the phrases “right upper semicontinuity”, “uniform convergence”, “Kirk asymptotic contractions”, and “Leader contraction”. It found the source and adjacent positive results but no prior counterexample to the two identified source claims. This supports only moderate novelty confidence; specialist review and a broader citation search remain appropriate.

6 Limitations and review recommendation

This counterexample is scoped to bounded-set uniform convergence. It is consistent with pointwise convergence, finite-time convergence of every individual orbit, existence and uniqueness of the fixed point, and the claim that the particular map is a Leader contraction. A reviewer should verify the source’s convention $n \geq 1$, the values of ϕ at r_k and at 1, and the exact conclusion of Corollary 3.9/Theorem 3.11. Subject to those checks, promotion as a valid counterexample is recommended.

References

- [1] H. Khandani, *A Study of Kirk’s Asymptotic Contractions via Leader Contractions*, arXiv:2507.05276v1, 2025.